

Eenheid 2.4

Note Title

(In Stewart is dit)
eenheid 2.5

2016/03/10

Kontinuiteit

Stewart pp 114 - 126
Studiegids p17

Definisie:

in Funksie f is kontinu in $x = \underline{a}$ as

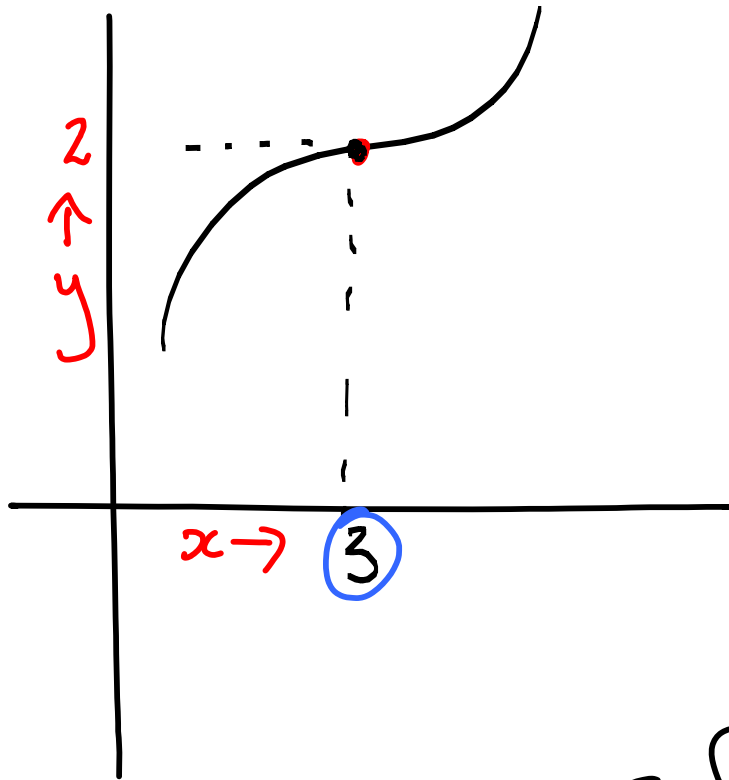
(i) $f(\underline{a})$ bestaan

(ii) $\lim_{x \rightarrow \underline{a}} f(x)$ bestaan

(iii) $\lim_{x \rightarrow \underline{a}} f(x) = f(\underline{a})$

Wat betekenen dit grafies??

f is continu in $x=3$.



f is "in een stuk"
"ononderbroke"
"aaneenlopend"

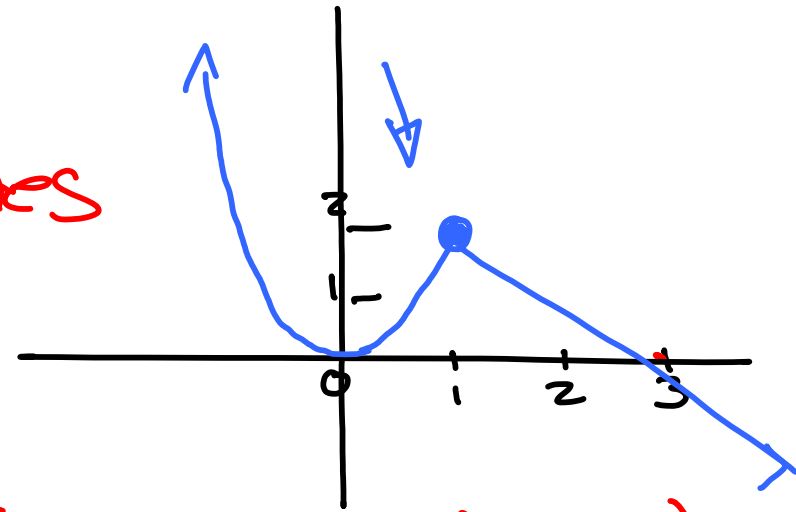
as f naby $x=3$ is

Voorbeeld: ① $f(x) = \begin{cases} 2x^2 & \text{as } x < 1 \\ -x+3 & \text{as } x \geq 1 \end{cases}$

Is f continu in $x=1$?

✓ kyk na al drie voorwaardes

① $f(1) = -1 + 3 = 2$



② $\lim_{x \rightarrow 1} f(x) = 2$ Want $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$

$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (-x+3) = 2$

$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x^2) = 2$

③ $f(1) = 2 = \lim_{x \rightarrow 1} f(x)$ Dus f is continu in $x=1$

~~$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$~~

$$\textcircled{2} \text{ Als } \underline{f(x)} = \begin{cases} kx+4 & \text{as } x < 2 \\ 5 & \text{as } \underline{x=2} \\ x^2+2k & \text{as } \underline{x} > 2, \end{cases}$$

bepaal k zodanig dat f continu is in $x=2$.

$$\textcircled{1} f(2) = 5$$

$$\textcircled{2} \lim_{x \rightarrow 2} f(x) \text{ bestaan as}$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$\text{Dus } 4+2k = 2k+4$$

$$\text{Dus } k \in \mathbb{R}$$

$$\textcircled{3} f \text{ is continu in } x=2 \text{ as}$$

$$f(2) = 5 = \lim_{x \rightarrow 2} f(x)$$

$$\text{Dan : } 5 = 4 + 2k \Rightarrow k = \frac{1}{2}$$

• Definisie: As f **nie** kontinu is in $x=a$ **nie**, sê ons f is **diskontinu** in $x=a$.
p115

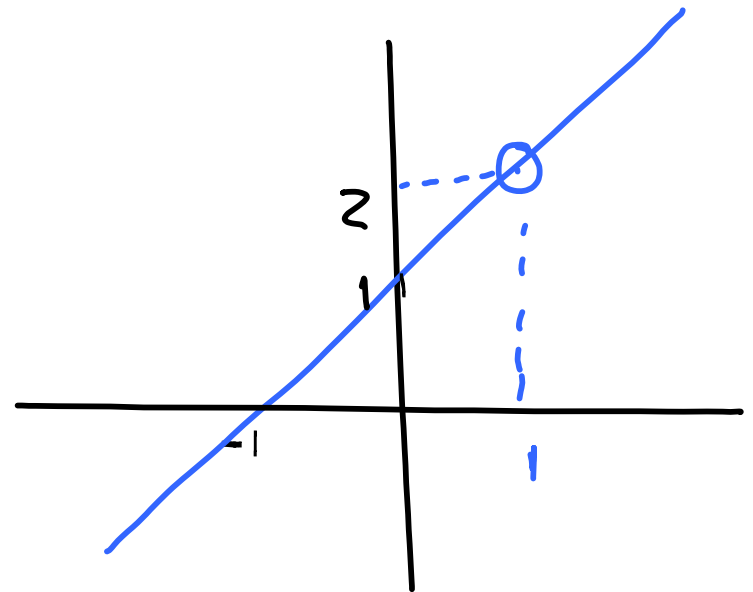
*3 Gevalle waar f diskontinu is.

① Verwyderbare Diskontinuiteit

Is $f(x) = \frac{x^2-1}{x-1}$ kontinu in $x=1$?

$$D_f = \mathbb{R} \setminus \{1\}$$

$$f(x) = \frac{\cancel{(x-1)}(x+1)}{\cancel{x-1}} = x+1$$

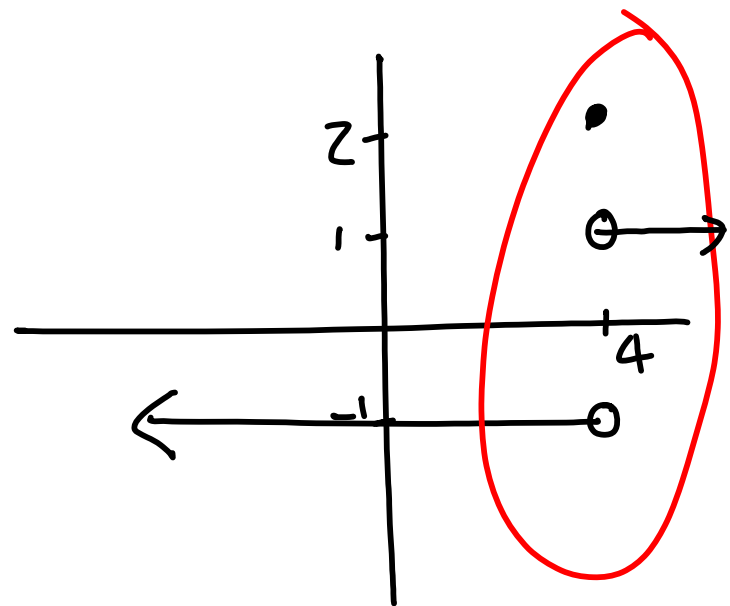


② Sprungdiskontinuität

Is $f(x) = \begin{cases} \frac{|x-4|}{x-4} & \text{as } x \neq 4 \\ 2 & \text{as } x = 4 \end{cases}$ kontinu in $x=4$?

$$= \begin{cases} \frac{x-4}{x-4} & \text{as } x > 4 \\ \frac{-(x-4)}{x-4} & \text{as } x < 4 \\ 2 & \text{as } x = 4 \end{cases}$$

$$= \begin{cases} 1 & \text{as } x > 4 \\ -1 & \text{as } x < 4 \\ 2 & \text{as } x = 4 \end{cases}$$



Toets die voorwaarden voor
continuïteit.

① $f(4) = 2$

② $\lim_{x \rightarrow 4} f(x)$ bestaan nie

Want $\lim_{x \rightarrow 4^+} f(x) = 1 \neq$

$\lim_{x \rightarrow 4^-} f(x) = -1$

f is diskontinu in $x = 4$

want $\lim_{x \rightarrow 4} f(x)$ bestaan nie

SPRONG Diskontinuitéit

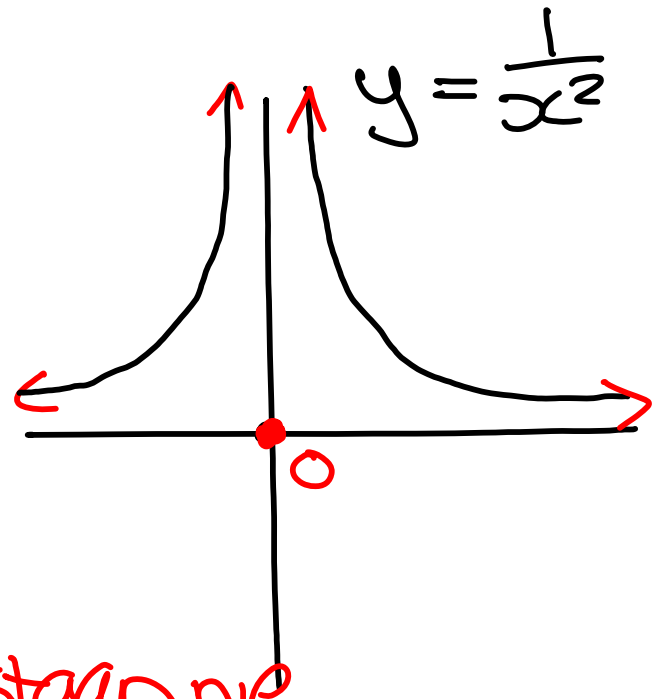
③ Oneindige diskontinuiteit

Is $f(x) = \begin{cases} \frac{1}{x^2} & \text{as } x \neq 0 \\ 0 & \text{as } x = 0 \end{cases}$ kontinu in $x=0$?

Toets die voorwaardes vir kontinuïteit!

✓ ① $f(0) = 0$

② $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty \quad \therefore \lim_{x \rightarrow 0} \frac{1}{x^2}$ bestaan nie



✗ ③ Dus f is diskontinu in $x=0$ want
 $\lim_{x \rightarrow 0} \frac{1}{x^2}$ bestaan nie

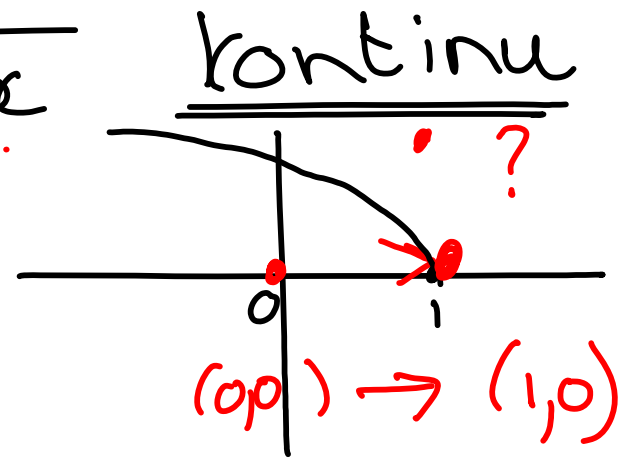
Stelling: p116 • in Functie f is kontinu
van regs in 'n punt a as
 $\lim_{x \rightarrow a^+} f(x) = f(a)$.

• in Functie f is kontinu
van links in 'n punt a as
 $\lim_{x \rightarrow a^-} f(x) = f(a)$

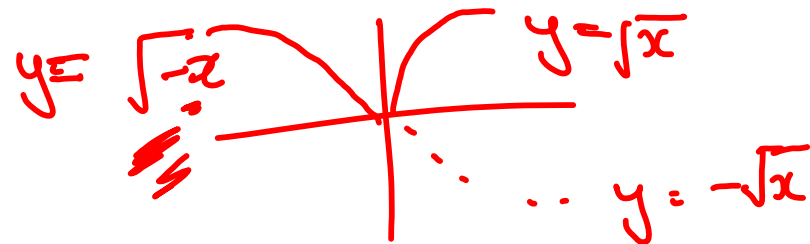
Voorbeeld:

Is die funksie • $f(x) = \sqrt{1-x}$

- ✓ ① van links af in $x=1$
- ② van regs af in $x=1$
- ③ in $x=1$



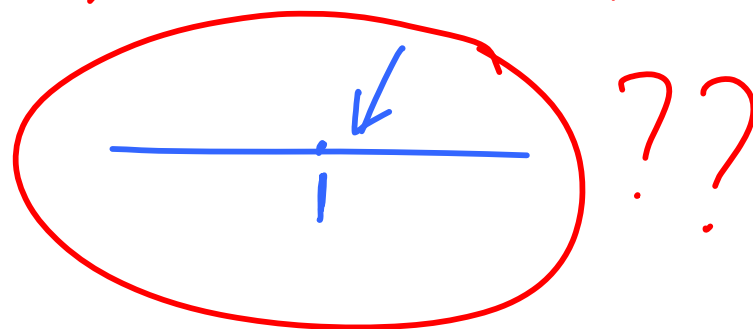
$$\textcircled{1} \checkmark f(1) = \sqrt{1-1} = 0$$



$$\checkmark \lim_{x \rightarrow 1^-} \sqrt{1-x} = 0$$

• Dus $f(1) = 0 = \lim_{x \rightarrow 1^-} f(x) \therefore f$ is links continu in $x=1$

$$\textcircled{2} \checkmark \underline{f(1)} = 0$$



$\times \lim_{x \rightarrow 1^+} \sqrt{1-x}$ bestaan nie

$\therefore f$ is nie regs kontinu in $x=1$

$$\textcircled{3} x=1 \quad ?$$

$$\checkmark f(1) = 0$$

• $\lim_{x \rightarrow 1} f(x)$ bestaan nie want $\lim_{x \rightarrow 1^+} f(x)$ bestaan nie
 f is dus diskontinu in $x=1$

- Kontinuiteit op in interval p 117
Definitie: Selfstudie
in functie is kontinu op in interval
as dit kontinu is in elke punt van die
interval.

Stelling 4 p 117 Kontinuiteit van kombinasies
van functies.

as f en g kontinu is in a en c is
konstantes, dan is die volgende funksies
ook kontinu in a :

1. $f + g$

2. $f - g$

3. $c \cdot f$

4. $f \cdot g$

• 5. $\frac{f}{g}$? as $g(a) \neq 0$

Stelling 5 p 118

Bv. $f(x) = x^3 + 2x^2 - x + 10$

* Enige polinoom is oral kontinu
dws. dit is kontinu op $\mathbb{R} = (-\infty, \infty)$

* Enige rasionale functie is kontinu
waar ook al dit gedefinieerd is;
dws. dit is kontinu in sy definisie-
verzameling

• $f(x) = \frac{x^3 + 3x^2 + x + 1}{(x-1)(x+2)}$ $D_f = \mathbb{R} \setminus \{1, -2\}$

Stelling 7 p 120

Die volgende funkties is kontinu
in elke punt van hulle definisie-
verzameling.

- polinome
 - rationale funksies
 - wortelfunksies
 - trigonometriese funksies
 - inverse trigonometriese funksies
 - eksponensiaal funksies
 - logaritmiese funksies
-

Voorbeelde: op watter intervale
is die volgende funksies kontinu?

① $f(x) = \frac{x-1}{x-2}$ Rationale funksie

f is kontinu op $D_f = \mathbb{R} \setminus \{2\}$

$$\textcircled{2} \quad g(x) = \frac{e^x}{\sqrt{x}} = \frac{f(x)}{h(x)}$$

apart $f(x) = e^x \Rightarrow D_f = \mathbb{R}$
 $h(x) = \sqrt{x} \quad D_h = [0, \infty)$

$$e^x: \quad D_f = \mathbb{R}$$

$$\sqrt{x}: \quad D_h = (0, \infty)$$

Dus g is kontinu op $(0, \infty)$

$\textcircled{3}$ Prakties no 45

$$f(x) = \begin{cases} cx^2 + 2x & \text{as } x < 2 \\ x^3 - cx & \text{as } x \geq 2 \end{cases}$$

Vir watter waardes van c (konstant)
 is die funksie f kontinu op $(-\infty, \infty)$?

Stelling 8 p 120 (geef aanleiding tot)

kontinuiteit van saamgestelde
funkties: " $\lim$ " $f(g(x)) = f(\lim_{x \rightarrow a} g(x))$

✓ Stelling 9 p 121

As g kontinu is by a en f is
kontinu by $g(a)$, dan is die
saamgestelde funksie $f \circ g$ gegee
dus $\underline{\underline{(f \circ g)(x) = f(g(x))}}$ kontinu by a

Voorbeeld : ① $h(x) = \sin\left(\frac{1}{x-2}\right) = f(\underline{g(x)})$

• $g(x) = \frac{1}{x-2}$ is continu op $\mathbb{R} \setminus \{2\} = D_g$

• $f(x) = \sin x$ is continu op $\mathbb{R}$

Dan $h(x) = \sin\left(\frac{1}{x-2}\right)$ is continu op

$$D_h = (-\infty, 2) \cup (2, \infty)$$

② $f(x) = \sqrt{x+2}$ en $g(x) = \frac{1}{x-2} \text{ (} x \neq 2 \text{)}$

• $(g \circ g)(x) = g\left(\frac{1}{x-2}\right)$

$= \frac{1}{\left(\frac{1}{x-2}\right) - 2}$

$= 1 \div \frac{5-2x}{x-2}$

$= \frac{1 - 2(x-2)}{x-2}$

$= 1 \times \frac{x-2}{5-2x}$

$= \frac{x-2}{5-2x}$

en $x \neq 2$

$x \neq 2$
en

$x \neq \frac{5}{2}$

$D_{g \circ g} = \mathbb{R} \setminus \left\{ 2, \frac{5}{2} \right\}$

$$\bullet (g \circ f)(x) = g(\sqrt{x+2}), \quad f(x) = \sqrt{x+2}, \quad g(x) = \frac{1}{x-2}$$

$$\underline{g(\sqrt{x+2})} = \frac{1}{\sqrt{x+2} - 2}$$

$$\sqrt{4} - 2 = 0$$

$$x+2 \geq 0$$

$$\Rightarrow x \geq -2$$

$$\text{or } \sqrt{x+2} - 2 \neq 0$$

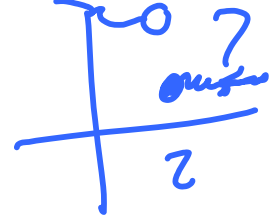
$$\Rightarrow (\sqrt{x+2})^2 \neq (2)^2$$

$$\Rightarrow x+2 \neq 4$$

$$\Rightarrow x \neq 2$$

$$D_{g \circ f} = [-2, 2) \cup (2, \infty)$$

45

$$f(x) = \begin{cases} cx^2 + 2x & \text{as } x < 2 \\ x^3 - cx & \text{as } x \geq 2 \end{cases}$$


$C = ?$ is f kontinu op $(-\infty, \infty) = \mathbb{R}$??

• f is kontinu op $(-\infty, 2)$ want $cx^2 + 2x$ polinoom

f is kontinu op $(2, \infty)$ want $x^3 - cx$ is polinoom

• In $x = 2$??

• $f(2) = ?$

• $\lim_{x \rightarrow 2} f(x) \checkmark$

$$\Rightarrow \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$C = ??$

Voltati self!

$$* \text{ Laat } f(x) = \begin{cases} 2a|x-3| & \text{as } x \geq 2 \\ 8 & \text{as } x = 2 \\ -x^2 + 4x + a & \text{as } x < 2 \end{cases}$$

Bepaal a zodat f continu is in $x=2$

$$• f(2) = 8$$

$$• \lim_{x \rightarrow 2^+} 2a|x-3| = 2a|-1| = (2a)(1) = \underline{2a}$$

$$\lim_{x \rightarrow 2^-} -x^2 + 4x + a = -4 + 8 + a = 4 + a$$

$$\therefore \lim_{x \rightarrow 2} f(x) \text{ bestaan as } 2a = 4 + a$$

$$a = 4$$

$$• \lim_{x \rightarrow 2^+} f(x) = 2(4) = 8 = \lim_{x \rightarrow 2^-} f(x) = 4 + 4 = 8$$

Dus f is continu in $x=2$ vir $a=4$

$$\text{Want } \lim_{x \rightarrow 2} f(x) = 8 = f(2)$$